



M-Tech Production & Marketing

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Week 4 - Day 2 Report

Internship Program — Batch 2026

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1. Introduction

Experiment No. 14 focused on studying Retrieval-Augmented Generation (RAG), an advanced Artificial Intelligence technique that combines information retrieval with Large Language Models (LLMs) to generate more accurate and context-aware responses. Instead of relying solely on the knowledge stored within a language model, RAG retrieves relevant information from external documents or databases before generating an answer. This experiment was conducted to understand the complete RAG workflow, including document retrieval, embeddings, vector databases, semantic search, and response generation using retrieved context.

2. Objectives

- Understand the concept and working principle of Retrieval-Augmented Generation (RAG).
 - Learn how external knowledge sources improve the performance of Large Language Models.
 - Study the process of document retrieval and context generation.
 - Explore embeddings and vector databases used in RAG systems.
 - Understand how retrieved information is combined with user queries to generate accurate responses.
 - Analyze practical applications of RAG in real-world AI solutions.
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3. Task Description

The experiment involved studying the architecture and workflow of a Retrieval-Augmented Generation system. Different stages of the RAG pipeline were explored, including document collection, text chunking, embedding generation, vector storage, similarity search, context retrieval, and response generation using a Large Language Model.

The experiment also examined how user queries are converted into vector representations and matched against stored document embeddings to retrieve the most relevant information. The retrieved context is then supplied to the language model, enabling it to produce responses that are both accurate and supported by external knowledge.

4. Work Done

The experiment began with understanding the limitations of traditional Large Language Models and the need for external knowledge retrieval. The complete RAG architecture was then studied to understand how retrieval and generation work together within a single AI pipeline.

Different concepts including text embeddings, semantic search, vector databases, and document chunking were explored in detail. Practical examples demonstrated how documents are indexed, converted into numerical vector representations, and searched efficiently based on semantic similarity instead of simple keyword matching.

The experiment also covered the interaction between retrieval systems and language models, showing how retrieved document passages are incorporated into prompts before response generation. This workflow improves factual accuracy while allowing AI systems to answer questions using organization-specific documents, research papers, manuals, and other private knowledge sources.

5. Challenges Faced

One of the main challenges was understanding how semantic search differs from traditional keyword-based searching. Learning how embeddings represent textual meaning in numerical form required careful study of vector representations and similarity calculations.

Another challenge involved understanding the complete retrieval pipeline, including document chunking, embedding generation, indexing, and context retrieval before language model inference. These concepts were better understood through repeated analysis of practical examples and workflow demonstrations.

6. Key Learnings

This experiment provided valuable knowledge of Retrieval-Augmented Generation and its importance in modern Artificial Intelligence applications. I learned how embeddings, vector databases, semantic search, and document retrieval improve the reliability and factual accuracy of AI-generated responses.

The experiment also strengthened my understanding of how Large Language Models can be combined with external knowledge sources to develop intelligent systems for customer support, enterprise search, educational assistants, document analysis, and knowledge management applications. Additionally, I gained practical insight into the overall architecture of modern RAG-based AI systems.

7. Conclusion

The Retrieval-Augmented Generation experiment successfully demonstrated how combining information retrieval with Large Language Models leads to more reliable and context-aware AI systems. By retrieving relevant information before generating responses, RAG significantly enhances response quality while reducing factual errors and outdated information.

Overall, this experiment expanded my understanding of modern AI architectures, semantic search techniques, vector databases, and intelligent document retrieval systems. The concepts learned provide a strong foundation for developing advanced AI applications capable of utilizing both language understanding and external knowledge sources effectively.

Intern Signature:



Supervisor:

Name:

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Name:

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Designation: